

SANJAY CHOUHAN

Full-Stack Backend Developer · Node.js · Real-Time Systems · MongoDB

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PROFESSIONAL SUMMARY

Backend developer with ~1.5 years of paid experience building production-grade systems in Node.js, Express, MongoDB, and Redis. Shipped a live real-time chat platform (WebSocket + WebRTC) serving concurrent users, and contributed backend architecture for a large-scale Hospital Management System covering ERP, billing, and authentication workflows. Experienced in JWT-based auth, Redis caching, queue-based background jobs, and aggregation pipeline optimization. Strong familiarity with Django/Python for secondary stack. All major projects are deployed and publicly accessible.

TECHNICAL SKILLS

Languages: JavaScript (Node.js), Python, SQL, HTML, CSS

Backend: Node.js, Express.js, Django, Django REST Framework, Flask

Databases & Cache: MongoDB, MySQL, PostgreSQL, Redis (pub/sub, caching, queues)

Real-Time: WebSockets (Socket.io), WebRTC (audio/video calling), Event-Driven Architecture

Auth & Security: JWT, Role-Based Access Control (RBAC), API Security, Session Management

Cloud & Storage: Cloudflare R2 (S3-compatible), REST API integration, Metered.ca TURN

Tooling: Git, GitHub, VS Code, Postman, Linux/Ubuntu, Background Jobs (cron)

ML (supplementary): Scikit-learn, Pandas, NumPy, Flask model serving

PROFESSIONAL EXPERIENCE

Backend Developer · MARCNC

Jun 2026 – Jul 15, 2026

Remote

Built and maintained backend services for a large-scale Hospital Management System spanning ERP, inventory, billing, reporting, user management, and healthcare workflows.

- Architected secure REST APIs in Node.js/Express for patient management, inventory tracking, reporting, and multi-role administration; system handles concurrent requests across multiple hospital departments.
- Implemented JWT authentication and granular role-based authorization (RBAC) covering 5+ user types including admin, billing, pharmacy, and clinical staff.
- Optimized MongoDB and SQL query performance through strategic indexing, aggregation pipelines, and schema redesign — reducing average query time on high-frequency billing routes by an estimated 40%.
- Integrated Redis caching and queue-based background processing to offload resource-intensive report generation and notification tasks from the main request thread.
- Integrated with ECHS government health scheme API, mapping complex service structures (roomChargesList, consultationList, expenseList) to internal billing data models.

Tech Stack: Node.js, Express.js, MongoDB, MySQL, Redis, JWT, REST APIs, Queue Systems

Full-Stack Development Instructor · MTB Pvt. Ltd

May 2025 – Jun 2026

- Delivered structured training on backend and full-stack development to 20+ students per cohort, covering Python, Django, Node.js, databases, and JavaScript frameworks.
- Designed hands-on project modules — each student shipped at least one deployed REST API before course completion.
- Maintained a 4.7/5.0 instructor rating across batches; mentored learners on debugging, Git workflows, and industry-standard code review practices.

Tech Stack: Python, Django, Node.js, Express.js, MySQL, PostgreSQL, HTML/CSS, JavaScript, Git

Backend Developer Intern · Pisoft Informatics Pvt. Ltd

Aug 2024 – Jan 2025

- Developed backend modules for e-commerce platform including cart, order processing, product catalog, and inventory management using Django and MySQL.
- Designed relational schemas for 8+ entities (products, variants, orders, customers, inventory, addresses, categories, admins), reducing JOIN complexity and improving query performance.

- Built and documented REST APIs consumed by a React frontend; reduced average API response time by ~30% through query optimization and selective indexing.

Tech Stack: Python, Django, MySQL, REST APIs, HTML/CSS, JavaScript

KEY PROJECTS

Buzz — Real-Time Chat Application → noira.sbs

Live · Ongoing

Production-deployed real-time messaging platform with one-to-one chat, media sharing, WebRTC audio/video calls, and advanced privacy controls.

- Engineered WebSocket (Socket.io) messaging with delivery/seen status sync, online presence tracking, real-time typing indicators, and a 30-second grace-period reconnection system to handle brief disconnects gracefully.
- Implemented WebRTC peer-to-peer audio/video calling with TURN server integration (Metered.ca), ICE candidate buffering, and mDNS candidate filtering for reliable call establishment across NAT environments.
- Built a background cron job for delivery/seen status reconciliation and auto-deletion of messages older than 30 minutes; eliminated ~15% stale-status accumulation during high-churn periods.
- Integrated Cloudflare R2 for media uploads with recovery workflows for deleted media references.

Tech Stack: Node.js, Express.js, Socket.io, WebRTC, MongoDB, Redis, Cloudflare R2, JWT

ecom_manager — E-Commerce Data Engine → backcrafter.shop

Live

Centralized dashboard engine that aggregates and manages e-commerce data using a custom storage layer built on Google Drive and the local filesystem — no traditional database required.

- Designed a custom file-based storage engine using Google Drive API and local filesystem as the data layer — storing both proprietary application data and client-supplied datasets without relying on a traditional database.
- Built multi-tenant authentication, secure data sync pipelines, and custom schema mapping to standardize product, order, and user records across client data sources.

Tech Stack: Node.js, Express.js, Google Drive API, Filesystem Storage, REST APIs, RBAC

Asset Infinity Clone — Full-Stack Asset Tracker → asset-infinity.netlify.app

Live

Complete frontend + backend replica of Asset Infinity enterprise tracking software.

- Designed multi-tier schema for hardware tracking, lifecycle logging, status indicators, and organizational asset allocation across departments.
- Implemented real-time dashboard analytics for inventory depletion and user utilization tracking.

Tech Stack: JavaScript, Node.js, Express.js, MongoDB, Tailwind CSS

Employee Salary Prediction System

- Built end-to-end ML pipeline: data preprocessing (Pandas/NumPy), Scikit-learn model training, and Flask-based model serving API.

Tech Stack: Python, Flask, Scikit-learn, Pandas, NumPy

EDUCATION

Diploma in Computer Engineering · G.D. Govt. Polytechnic, Hisar

2021 – 2024

CERTIFICATIONS

- Django Backend Development — 6-Month Internship Certificate, Pisoft Informatics Pvt. Ltd